

EDUCATION COMPETITION AND FERTILITY INTENTION: EVIDENCE FROM CHINA'S PRIVATE TUTORING BAN*

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Low fertility presents a major challenge for many nations. We study whether restricting competition-driven private tutoring can enhance fertility intentions by analysing China's 2021 private tutoring ban. Using nationwide surveys, we elicit respondents' fertility intentions under scenarios with and without the policy. The tutoring ban significantly increases expected total fertility by 7%–8%, with larger effects in cities under greater policy intensity. Decomposition reveals that the primary driver is perceived reduction in educational competition, followed by improved parental health, parent-child relationships, and reduced time and monetary costs. We also find consistent results on actual birth rates three years after the policy.

JEL codes: H52, I28, J13

Fertility is declining globally, with rates in more than half of all countries in 2021 below the replacement level (Bhattacharjee *et al.*, 2024). Coupled with increasing longevity, it leads to a demographic shift towards an ageing population, resulting in substantial socioeconomic repercussions (Eggertsson *et al.*, 2019).¹ In response, many countries have implemented pro-fertility initiatives, such as daycare subsidies, tax incentives for families with multiple children and extended paid maternity leave (Zhang *et al.*, 2021). Despite these efforts, most countries still struggle to stop the decline in fertility rates, much less reverse it. Even when some of these pro-natal policies show effectiveness, the impact is often small relative to the substantial government expenditure on these programmes, which can amount to 1% to 4% of GDP (Laroque and Salanié, 2004; Cohen *et al.*, 2013; Tomáš *et al.*, 2019).

Theoretically, low fertility can be directly linked to the high costs of child-rearing (Becker and Barro, 1988; Becker *et al.*, 1990). As socioeconomic development progresses, increased return to human capital leads to a rising intensity of parenting and greater investment in child education in many countries (Doepke and Zilibotti, 2019). Moreover, limited seats in elite universities often induce competitive motives in educational investment to obtain higher relative rankings, a trend observed in both Western (Ramey and Ramey, 2010) and Asian societies (Guo and Qu, 2022; Kim

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¹ These include and are not limited to a contraction of the labour force (Peters and Walsh, 2021; Hopenhayn *et al.*, 2022; Karahan *et al.*, 2024), increased pressure on social support systems for the elderly (Attanasio *et al.*, 2007; Coeurdacier *et al.*, 2014) and a potential slowdown in innovation (Liang *et al.*, 2018; Jones, 2022).

et al., 2024). Doepke and Zilibotti (2019) explained that competitive pressure can trickle down to earlier stages of education, leading to high anxiety among parents and prompting significant investment in exam tutoring, as observed in countries like the United States and Switzerland. In Asian countries, the phenomenon of ‘education fever’ is also been evident for a long time.² Educational competition may, not only lead to over-investment (Gu and Zhang, 2024), but also create high psychological pressure for both parents and children (Chen *et al.*, 2021). This may partly explain why many childbirth subsidy policies are less effective in boosting fertility, because parents will still compete through private investment and incur additional non-pecuniary costs such as time commitment and mental pressure.

A natural question thus arises: will attempts to restrict private tutoring and reduce educational competition effectively increase the willingness to have more children? To answer this question, we examine the private tutoring ban implemented by the Chinese government in July 2021. This policy seeks to ease the ‘education fever’ by curbing the rapidly growing for-profit private tutoring sector, which has placed substantial financial and time burdens on parents (Wei, 2024), while also raising concerns about its impact on students’ physical and mental health (Ren *et al.*, 2017; Jiang *et al.*, 2021).

Taking this sudden policy as a natural experiment, we evaluate whether and how the private tutoring ban affects fertility intentions. We first conducted a within-subject survey one semester after the policy implementation on a national educational cloud platform, obtaining nearly 1,200 parents who were directly affected by the policy shock. We began by eliciting parents’ subjective beliefs about the probability of having an additional child under two scenarios—with and without the private tutoring ban—based on the notion that properly elicited intentions have been shown in the literature to be an effective predictor of actual choices (Fuster and Zafar, 2023).³ We then collected detailed retrospective information related to actual child education, including monetary and time costs, parental health and parent-child relationships, for the semester both before and after the implementation of the tutoring ban.

Using both a family fixed-effect model and a difference-in-differences framework, we find that, under the scenario with the tutoring ban, the subjective probabilities of having a second child and a third child increased to 32.5 and 12.9 percentage points, respectively—a 41.9% and 56.2% increase relative to the sample mean under the scenario without the tutoring ban. This change in fertility intention, if fully fulfilled, is equivalent to an increase in the total fertility rate (TFR) by 7%. This considerable increase in fertility intention is noteworthy, as cross-country evidence indicates that achieving a similar TFR increase typically requires a country to spend 1% of its GDP on pro-natal subsidies (Liang, 2018). The tutoring ban’s effect on fertility intentions exhibits heterogeneity, with a significantly smaller impact on mothers compared to fathers, and a significantly reduced effect for respondents with a bachelor’s degree or higher. We further find that the greater the monetary and time investments parents make prior to the policy, the more their fertility intention increases following the tutoring ban.

To mitigate concerns about the experimenter demand effect, self-selection in survey participation and the potential impact of the COVID-19 pandemic, we also conducted a supplementary between-subject survey with 1,348 participants using the panel service of a leading professional

² For instance, in South Korea, 75% of children participate in after-school programmes, with families spending 9.2% of their income per child on extracurricular education (Kim *et al.*, 2024). Among urban Chinese families, these numbers are 24.4% and 11.2%, respectively (Wei, 2024).

³ The recent fast-growing literature has shown that subjective beliefs are important in understanding actual decisions, including important life decisions such as marriage and fertility (Wiswall and Zafar, 2021; Gong *et al.*, 2025), retirement (Giustinelli and Shapiro, 2024) and savings and investment (D’Acunto and Weber, 2024).

survey company at the end of 2024. We find a robust and similar-sized 8% increase in expected total fertility across survey designs and samples. The fact that we find similar policy impacts on intended fertility in both the within- and between-subject designs suggests that our original results are unlikely to be driven by the above concerns.

We then explore potential contributing factors for the increased fertility intentions. We first show that parents perceive a less competitive educational environment when the tutoring ban is implemented. In the main within-subject survey, 45% of the respondents think that *other* parents in the same grade will reduce private tutoring after the policy. In the supplementary between-subject survey, we further directly elicit parental belief on how competitive the child's education will be under each scenario using a ten-point scale. The average level of competition is perceived to be quite high at 8.3 without the policy and is substantially reduced by 44% in the scenario that implements the tutoring ban. In addition to lower perceived competition, we also find concrete evidence that the tutoring ban effectively alleviated educational costs in multiple dimensions, including significantly lower monetary expenditure on tutoring and parental time on their children's studies, as well as improved parent mental health and parent-child relationships.

We conduct a decomposition analysis to disentangle the relative contribution of the above-mentioned potential factors to the increased fertility intention. Together, these contributing factors account for more than 43% of the tutoring ban's effect. Among the explained variation, reduction in perceived competition accounts for the largest portion of the tutoring ban's positive effect on fertility intentions, followed by improved parental health, and then reductions in time and monetary costs. These findings underscore the critical role of reduced competition and non-pecuniary benefits—such as improved health and parent-child relationships—relative to direct monetary costs in shaping fertility decisions.

Lastly, we explore the short-run impact of the tutoring ban on actual fertility after three years of implementation, leveraging city-level variations in the exposure to the sudden policy. While the absolute level of fertility is dropping rapidly in China, we collect data on the city-level annual birth rates for periods both before (2016–20) and after the policy (2023–4) and find that the birth rates are declining significantly less in cities with higher policy intensity after the tutoring ban relative to those with lower intensity. A one-SD increase in policy intensity translates to a 7.0% slowdown in birth rate decline between 2020 and 2023. This finding of the policy impact on actual birth rates is broadly consistent with the higher fertility intentions that we elicited from the survey data.

This paper is related to several strands of literature. First, it contributes to the literature on the design and effectiveness of family policies in boosting fertility (Olivetti and Petrongolo, 2017). Existing research has examined the effect of the cash baby bonus (Laroque and Salanié, 2014; Zhou, 2022), public provision or subsidy of childcare (Bauernschuster *et al.*, 2016; Olivetti and Petrongolo, 2017), parental leave benefit (Lalive and Zweimüller, 2009; Raute, 2019), taxation incentive (Milligan, 2005; Eckstein *et al.*, 2019) and access and cost of public school (Aaronson *et al.*, 2014; Burlando and Bbaale, 2022). In a recent work, Kim *et al.* (2024) incorporated the competitive motive in a quantitative macroeconomic model and found that fertility would be substantially higher in a counterfactual world without this motive. Compared to these studies, as far as we know, our paper is the first to empirically examine the effect of the tutoring ban on fertility intentions using survey evidence, as well as actual fertility, an important and policy-relevant topic.

Second, this paper adds to studies on the relationship between education and fertility and the large literature on the quantity-quality trade-off (see a recent review by Guo *et al.*, 2022).

Most of the literature focuses on how the increased education of the mother herself affects her fertility (Becker *et al.*, 2013; Hazan and Zoabi, 2015; Fort *et al.*, 2016), and how the fertility level affects parental investment in child quality (Black *et al.*, 2005; Rosenzweig and Zhang, 2009; Wang and Zhang, 2018). One exception is Becker *et al.* (2012), who found a negative effect of children's education investment on parental fertility in 1816 Prussia. Our study contributes to this literature by focusing on whether lowering the competition and cost of children's education, fixing parental education, can increase fertility intention in a modern economy with a large population, but rapidly decreasing fertility.

Third, this study builds on the fast-growing literature on subjective expectations (see the recent handbook edited by Bachmann *et al.*, 2023). The properly elicited expectations and intentions are shown to be meaningfully correlated with actual choices in various contexts, such as fertility (Gong *et al.*, 2025), education (Jensen, 2010), job search (Mueller *et al.*, 2021) and homeownership (Bottan and Perez-Truglia, 2025). Existing studies have also examined how expectations and intentions adjust under different hypothetical scenarios and link to actual future choices when a particular scenario is realised. For example, Wiswall and Zafar (2021) found that the expected labour supply, earnings and spouse type, conditional on choosing a particular college major, are remarkably correlated with realised outcomes six years later. Giustinelli and Shapiro (2024) showed that the elicited work expectations across alternative health states predict retirement decisions two to four years later upon realised health shocks. Our paper adds to this literature by examining both intended and actual fertility—an important life decision with substantial policy relevance—under an education policy shock, and by providing evidence on their consistency.

Lastly, several recent studies have examined the immediate impact of the tutoring ban and found a decrease in tutoring expenditure and time (Liu *et al.*, 2024; Zhao *et al.*, 2024) and an improvement in student health measured by insomnia symptoms and pressure (Zhang *et al.*, 2024). Compared to these studies, our novelty is to use nationwide surveys to examine the effect of the tutoring ban on fertility intentions. Additionally, we provide unique data and evidence on the effects of the policy on perceived educational competition, parental health and parent-child relationships, dimensions that have not been examined in previous work. Our novel decomposition analysis also reveals that improvements in these aspects, particularly reductions in perceived educational competition, play a more significant role in boosting fertility intentions than monetary and time costs.

The remainder of the paper proceeds as follows. Section 1 introduces the institutional background. Section 2 describes the survey design and sample. Sections 3 and 4 present the empirical analyses. Section 5 concludes the paper.

1. Background

Private tutoring is prevalent in many places around the world, including the United States, several European countries and Asia, as documented by Doepke and Zilibotti (2019). An important contributing factor is the limited and highly differentiated resources in the educational system, especially higher education. Elite universities have a limited number of slots, and therefore admission is very selective. For example, gaining access to elite universities in the United States requires, not only near perfect high school grades and SAT scores, but also an outstanding portfolio of extracurricular activities, which have to be invested in and accumulated from earlier grades. In Switzerland, only about 20% of students can pass a high-stakes exam to enter the more academic track of high school, which paves the way to university. In both China and Korea,

college admission is primarily determined by a high-stakes centralised entrance exam. All of these factors lead to competition in educational investment for higher relative rankings so that students can stand out during examinations and admissions. This phenomenon is described as ‘the rug rat race’ by Ramey and Ramey (2010), and is formally analysed in quantitative models by Kim *et al.* (2024) and Gu and Zhang (2024). Parents start investing in private tutoring and other extracurricular coaching even at the very early stage of education (Chi and Qian, 2016; Doepke and Zilibotti, 2019; Wei, 2024), and, as shown by Kim *et al.* (2024), this status competition has spillover effects, prompting other parents to invest as well, even if they did not intend to initially.

Take China as a concrete example. The country provides nine years of compulsory education—six years of primary school and three years of middle school—during which the vast majority of students attend public schools. Private schools account for a small share of total enrolment, and most of them are typically geared toward students planning to study abroad.⁴ As a result, nearly all students aiming for admission to top domestic universities are enrolled in the public system. Admission to college is primarily determined by performance on the National College Entrance Examination (*Gaokao*), a high-stakes and highly competitive test taken at the end of high school (Jia *et al.*, 2025). Since *Gaokao* outcomes depend heavily on the quality of high school education, access to elite public high schools is highly sought after. Because high school admission is merit based and seats are limited, competition for these high schools is fierce, generating substantial academic pressure. This pressure does not begin in high school; rather, it cascades backward through middle school and even into primary school, as families invest heavily, particularly in after-school tutoring, from an early stage to keep their children on a competitive academic track.

Competitive motives have driven education expenditure among many households to escalating levels, making it one of the main costs of raising children. According to a nationally representative survey of 40,000 households in China in 2019 (Wei, 2024), although compulsory education is tuition-free and government funded, 24.4% of students participate in academic tutoring programmes. On average, families spend 11.2% of their income per child on these extracurricular education activities. This educational burden is disproportionately heavier in urban areas, where the participation rate and expenditure share are four and three times higher, respectively, than those in rural areas. Rapid expansion and rising demand for tutoring services have attracted a multitude of institutions and firms to this burgeoning industry. The tutoring sector has experienced swift growth, becoming significantly lucrative. By 2020, the industry valuation reached approximately \$120 billion, accounting for 0.8% of China’s GDP. The fierce competition between these firms has spurred intensive marketing campaigns, further exacerbating educational competition, and increasing anxiety among households about their children’s academic success.

The intense competition in education in China, not only imposes significant financial burdens on households, but also contributes to a variety of health and psychological problems among adolescents. A pervasive lack of sleep has become increasingly common among children, driven by time-consuming assignments after hours spent in tutoring institutions (Zhang *et al.*, 2023). In addition, there is evidence linking high obesity rates with excessive homework and insufficient physical activity (Ren *et al.*, 2017). Moreover, the high and sustained levels of stress experienced by students lead to depression, making suicide the leading cause of death among Chinese teenagers (Jiang *et al.*, 2021).

⁴ According to the Statistical Report on China’s Educational Achievements in 2022 released by the Ministry of Education, only 3.9% of students at primary and middle school were enrolled in private schools (http://en.moe.gov.cn/documents/reports/202304/t20230403_1054100.html, accessed July 2025).

These issues have sparked extensive social debates and drawn significant policy attention, culminating in the implementation of the private tutoring ban policy by the Chinese central government on 24 July 2021. This policy aimed to reduce the burden of homework and after-school tutoring for primary and middle school students (grades 1 to 9) and to alleviate the financial pressures on families concerning educational spending (Xue and Li, 2023).

The most important aspect of the policy is to effectively ban the for-profit after-school tutoring industry through several strict regulations (hereafter referred to as the ‘tutoring ban’). These regulations include the following: (a) existing tutoring institutions must convert into non-profit organisations, while the government stops issuing operating licences for new institutions; (b) capital investment and financing are strictly restricted—for example, tutoring companies are prohibited from being publicly listed on the domestic stock exchange; (c) after-school tutoring for compulsory education students is not allowed to cover core academic curriculum and cannot operate on weekends or holidays. In general, the tutoring industry substantially shrunk after the ban. At the end of the first semester after the policy, the Ministry of Education reported that the number of in-person tutoring institutions decreased by 83.8%, while online tutoring decreased by 84.1%.⁵ Huang *et al.* (2025) found that within four months of the tutoring ban, online job postings for tutoring-related firms decreased by 89%, and the number of tutoring-related firm exits tripled. It is worth noting that these policies do not affect in-school education expenditures, as compulsory education in China is free. However, they may impact a child’s study time and parental time if supervision is required.⁶

Finally, we briefly outline the fertility situation in China. According to the World Bank classification, the TFR in upper-middle-income countries declined from 1.96 in 2017 to 1.59 in 2022. Within this group, China’s TFR drop has been particularly striking, falling from 1.81 in 2017 to 1.18 in 2022 (United Nations, 2022). China now joins Japan (TFR = 1.3) and South Korea (TFR = 0.8) in forming a global ‘fertility valley’, representing some of the lowest birth rates worldwide. The rapid fertility decline has accelerated China’s transition to an ageing population, posing significant socioeconomic challenges for future growth. As a result, how to raise fertility has attracted substantial attention and become a key policy priority for the government.

2. Survey and Sample Description

2.1. *The Survey*

We designed and administered a nationwide online survey to Chinese parents with children of relevant ages in early 2022 to collect detailed information on fertility intention and family education investment behaviours before and after the tutoring ban. In the following, we describe the survey device, sampling criteria and details of the questionnaire.

2.1.1. *The sample*

In the main survey, we target parents with at least one child attending primary or middle school, who were directly affected by the policy shock. We aim to collect information covering two entire

⁵ Data from the Ministry of Education of China: <https://www.globaltimes.cn/page/202112/1242974.shtml> (retrieved 16 October 2025).

⁶ The policy also establishes requirements to reduce the in-school burden, including restrictions on homework and school hours. For instance, written homework is prohibited for grades 1 and 2, while the maximum allowed time for written homework is 60 minutes for grades 3 to 6 and 90 minutes for grades 7 to 9. It also addressed the misalignment between school hours and parents’ work schedules by postponing school start times and requiring schools to provide after-school non-academic sessions for students in need.

semesters before and after the implementation of the tutoring ban (that is, spring 2021 and fall 2021). Therefore, we further exclude parents whose children started primary school only after the tutoring ban, or those whose children transitioned from primary to middle school or from middle to high school, because such educational transitions are substantial, and thus make the two semesters before and after the policy not comparable.

To achieve a nationwide sample of parents, we distribute survey questionnaires on a national educational cloud platform. The platform was officially launched in 2015 following the plan by the Ministry of Education of China to provide primary and secondary schools in all provinces with an integrated intelligent education solution for instruction, testing and management. At the time of the survey distribution, the school coverage in each province is approximately 55%. In daily use, teachers, students and parents use their respective accounts to log into the platform system via computers or mobile devices. We distributed the online survey for two weeks in early spring 2022, one full semester after the implementation of the tutoring ban. After logging in, eligible parents who meet our sampling requirement see a clear advertisement banner on the front page to invite them to participate in a paid survey related to the tutoring ban. Participants enter the survey page by clicking the banner. After completion of the whole survey, which takes on average 5–8 minutes, we pay each participant a random amount between 10 and 20 RMB (1.5 to 3 USD).

2.1.2. Survey design

At the beginning of the questionnaire, we collect basic information about the family, including the age, education, Hukou (the household registration system) and marital status of both parents, as well as the gender, age and school grade of all children. After this, there are two main survey modules. The first is to elicit the fertility intention of parents, and the second module collects detailed information on child education. See [Online Appendix D](#) for the questionnaire.

Fertility intention elicitation: We follow the fast-growing literature on survey-based subjective belief to elicit parents' beliefs about the probability of having an additional child under two scenarios in a within-subject design (Fuster and Zafar, 2023). The first scenario is one where the policy does not exist, and the other scenario is where the policy is implemented. We do not specify any other conditions or assumptions. One potential concern of the within-subject design in this case is the experimenter demand effect. To mitigate this concern, we additionally run a supplementary between-subject survey and find robust results, which are discussed in Section 4.2. The questions (translated from Chinese) are given as follows.

[* The unit is %. Please answer in integers between 0-100 in each box. *]		
	Tutoring ban not implemented	Tutoring ban implemented
<i>(For parents currently with one child)</i>		
— Likelihood of having the 2 nd child:	[]	[]
— Suppose you already have two children, likelihood of having the 3 rd child:	[]	[]
<i>(For parents currently with two children)</i>		
— Likelihood of having the 3 rd child:	[]	[]

Respondents are asked to provide a probability between 0% and 100% of having a second and third child (for parents with one child) or a third child (for parents with one or two children) in the future, under each corresponding scenario. Eliciting beliefs using probabilities allows respondents to express belief uncertainty, following common practice in this literature (Wiswall and Zafar, 2021). In our setting, for parents who do not have a decisive idea about having another child—a situation likely for many—probability-based questions may be easier to answer than forcing a binary choice, which would require rounding their belief to 0 or 1 in an arbitrary way. For respondents with firm decisions, 0% or 100% remains a valid option. We can then calculate the expected total number of children for each family as a proxy for total fertility using these subjective probabilities.

Child education and health: After eliciting fertility intentions, in the second module, we collect detailed retrospective information on child education investment and health-related status one semester before and after the implementation of the policy. Following common practice in household surveys, we design the questions using two different methods based on the nature of the variables. First, for variables that can be measured objectively in natural units, we collect them for two semesters—before and after the tutoring ban implementation—separately. These variables include time and monetary investments related to education, such as school start and end times, time and money spent on academic tutoring, parental time spent on a child's study and total monthly education expenditure.

The second type of variable pertains to more subjective perceptions, such as self-rated physical and mental health, and parent-child relationships. Given the inherent challenges in objective quantification, we employ a ten-point Likert scale to directly elicit parental responses on the magnitude of the *change* following the tutoring ban, compared to the pre-policy period, with zero points indicating a significant deterioration, five points for no change and ten points representing a substantial improvement.

2.2. Summary Statistics

We receive a sample of parents from 1,154 families with primary and middle school students. They are located in 136 cities in twenty-eight of the thirty-one Chinese provinces and are roughly equally distributed among school grades. Panel A of Table 1 summarises the background information of the parent respondent. The average number of existing children in our sample is 1.5, similar to the national average of 1.43 for the same parental age range in 2020.⁷ Given the average number of existing children, the average monthly family income of 10,507 RMB translates into an annual income of 36,014 RMB per capita, which is very close to the national disposable income per capita of 35,128 RMB in 2021. In addition, 52.3% of parents have urban Hukou, also close to the national statistic of 47.3%.⁸

Panel B of Table 1 summarises the family's private education investment during the baseline semester before the tutoring ban was implemented. On average, each student spends 2.2 hours per week on private tutoring for the academic curriculum, with a cost of 795 RMB per month. They additionally spend similar amounts of time and money on tutoring for non-academic classes such as art or sports. In total, parents spend 26.2% of their monthly family income on all the

⁷ Data from the 2020 round of the China Families Panel Study, a nationally representative longitudinal survey in China.

⁸ Data from the National Bureau of Statistics of China: <https://data.stats.gov.cn/english/>.

Table 1. *Summary Statistics.*

Variable	Obs.	Mean	SD	Min	Max
<i>Panel A: family information</i>					
Mother's age	1,154	38.832	6.880	25	55
Hukou status (urban = 1, rural = 0)	1,154	0.523	0.500	0	1
Marriage status (married = 1, divorced = 0)	1,154	0.964	0.187	0	1
Highest education of parents (year)	1,154	13.786	3.132	9	19
Number of existing children	1,154	1.501	0.628	1	5
Monthly family income (1,000 RMB)	1,105	10.507	12.237	1	100
<i>Panel B: family education investment (baseline semester before tutoring ban)</i>					
Private academic tutoring (hour/week)	1,138	2.207	3.679	0	40
Expenditure on academic tutoring (1,000 RMB/month)	1,064	0.795	1.632	0	20
Total education expenditure (1,000 RMB/month)	1,105	2.618	4.397	0	58
Total education expenditure (share of income)	1,105	0.264	0.201	0	1
Parental time spent on supervising child's study (hour/day)	1,022	1.363	0.902	0	6

Notes: One parent per family was surveyed.

education-related activities of their children, close to the national average of 28.8% reported by Wei (2024). In addition, parents spend roughly 1.4 hours a day supervising their children's studies. Together, these illustrate the high cost of child education during the primary and middle school stages, both in terms of money and parental time.

3. Main Results

3.1. Fertility Intention

We first report raw data on elicited parental beliefs about the probability of having an additional child. Panels (a)–(c) of Figure 1 plot the average subjective probability of having the next child in each of the two scenarios, without and with the implementation of the tutoring ban. For parents currently with one child, the probabilities of having a second and third child when there is no tutoring ban are 22.9 and 8.3 percentage points, respectively. With the tutoring ban implemented, these probabilities increase substantially to 32.5 and 12.9 percentage points, or by 41.9% and 56.2%, respectively. A similarly large effect is found for the probability that parents with two children will have a third child, which doubles from the baseline of 9.1 percentage points without the policy.

In Figure 1(d), we further calculate the expected total number of children for each family as a proxy for total fertility by combining the number of their existing children and integrating the conditional probabilities of having each additional child. For example, for parents with one child and the reported probabilities of having the second and third children being p_2 and p_3 , the expected total number of children is given by $(1 + p_2 + p_2p_3)$.⁹ We compare the expected total number of children for each family and find that the expected fertility increases from 1.610 to 1.726 under the situation with the tutoring ban, a meaningful step toward the ideal number of children of 1.86 reported in a nationally representative sample in 2018 (Yu and Xie, 2021).

⁹ Having four or more children is rare in China (less than 1.6% of all mothers according to the 2015 census). To ease the cognitive load of the survey participants, we did not elicit the belief of having higher-order children beyond three in the survey, and assumed this away in the calculation.

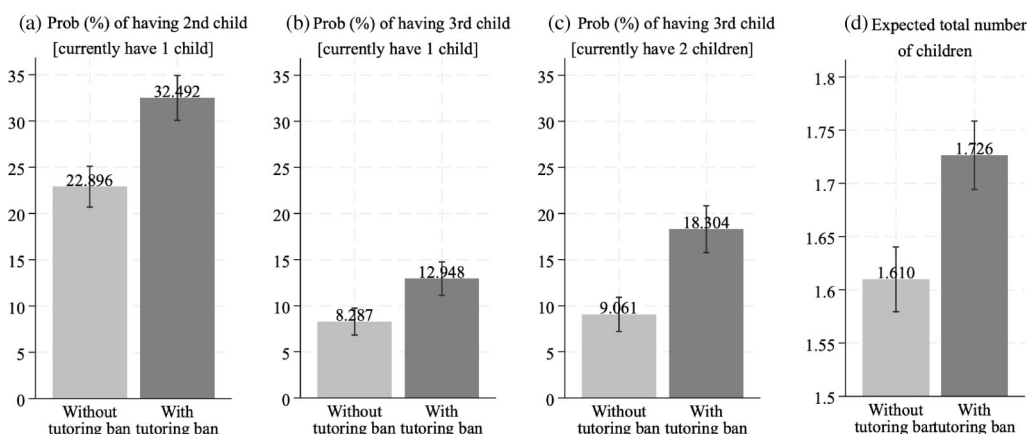


Fig. 1. Raw Comparison of Fertility Intention under Different Scenarios.

Notes: Each bar represents the mean with the 95% confidence interval. Regarding the samples, (a) and (b) include parents who currently have one child, (c) includes parents who currently have two children and (d) includes all parents.

Using the within-subject design of eliciting parental beliefs on fertility, we then quantify the above policy effects using a family fixed-effect model:

$$Y_{is} = \beta_0 + \beta_1 Ban_s + \delta_i + \varepsilon_{is}. \quad (1)$$

Here Y_{is} includes the probabilities of having an additional child, conditional on the number of existing children, as well as the integrated expected total number of children, of family i under scenario s ; Ban_s is a binary indicator for implementing the tutoring ban in scenario s ; δ_i is the family fixed effect and ε_{is} is the residual term.¹⁰

Regression results in Table 2 confirm that the positive effects of the tutoring ban on fertility intentions found in Figure 1 are all statistically significant. Controlling for family fixed effects, the expected total fertility significantly increases by 0.109 children (or 6.8%) with the tutoring ban. Cross-country evidence indicates that such a 0.1 increase in the TFR typically requires a country to spend 1% of its GDP on pro-natal subsidies (Liang, 2018). Stone (2020) also found that increasing child benefits by 10% of household income leads to a 0.5%–4.1% increase in birth rates, summarising existing policy evaluations. Therefore, restricting private tutoring and competition can potentially achieve an effect size in boosting fertility that is equivalent to a large government fiscal expenditure.

We have performed several robustness checks. The first concerns the potential effects of COVID-19. The tutoring ban was implemented during the COVID-19 pandemic. However, we show that during the studied period, the number of COVID-19 cases remained stable at a very low level and most educational and daily activities proceeded as normal, as detailed in Online Appendix B. To further account for potential regional variations in occasional outbreaks and the progression of the pandemic, we add three additional control variables to the regression at the

¹⁰ SEs are clustered at the province level. Because the survey was distributed on each of the provincial platforms serving schools and parents in the corresponding province, this forms a sampling design problem, as formally discussed by Abadie *et al.* (2023), and requires the SE adjustment to be the same at the province level in addition to accounting for the potential within-region correlation between residuals.

Table 2. *Fertility Intention.*

Dep. var.:	Probability (0%–100%) of having			
	a 2 nd child (have 1 child)	a 3 rd child (have 1 child)	a 3 rd child (have 2 children)	Expected number of children (all)
Dep. var. mean:	[22.896] (1)	[8.287] (2)	[9.061] (3)	[1.610] (4)
Tutoring ban	9.474*** (0.561)	4.221*** (0.704)	9.104*** (1.104)	0.109*** (0.007)
Family FEs	Yes	Yes	Yes	Yes
N	1,212	1,206	864	2,070

Notes: In columns (1)–(3), the dependent variable is the probability (0%–100%) of having a second and third child (for parents currently with one child) and a third child (for parents currently with two children), respectively. In column (4), the dependent variable is the expected total number of children (which uses the full sample of parents). The main independent variable is an indicator for the scenario where the tutoring ban is implemented, and all regressions control for family fixed effects. Regarding the samples, columns (1)–(2) include parents who currently have one child, column (3) includes parents who currently have two children and column (4) includes all parents. The number of observations slightly differs between columns (1)–(2) and between column (4) and the sum of columns (1) and (3) due to missing responses in the corresponding dependent variable. SEs are clustered at the province level. Significance level: *** $p < .01$.

city-by-semester level, including the total number, volatility and growth rate of COVID-19 cases (Hu *et al.*, 2020). These variables, not only absorb the impact of the pandemic on the education system, but also, to a certain extent, control other unobserved factors such as income or fertility expectations. We find that fertility intention, as well as educational investment, are not sensitive to these controls in [Online Appendix Table A1](#).

The second robustness check concerns regional representativeness. The provincial distribution of this sample and the actual population from the 2020 census are compared in [Online Appendix Figure A2\(a\)](#). Our sample exhibits higher representation in certain provinces and lower representation in others, likely due to varying levels of platform usage across regions. To ensure that our findings are not driven by regional representation biases, we computed ex post sampling weights based on actual provincial population shares or the national share of urban Hukou. As shown in [Online Appendix Table A2](#), our results remain robust after adjusting for these sampling weights.¹¹

3.2. *Heterogeneity*

Next, we examine the heterogeneous effect of the tutoring ban on the expected number of children by interacting the *Tutoring ban* indicator in (1) with the corresponding group indicators of the respondents and report the results in Table 3. First, although we find that the expected number of children in the benchmark scenario without the tutoring ban does not differ by parent gender, the positive effect of the tutoring ban on fertility intention significantly smaller for mother: 0.04 lower than for fathers, whose effect is 0.13 (column (1)). It is also 0.03 smaller for respondents with a bachelor’s or higher degree (column (2)). The policy effect does not significantly decrease

¹¹ The between-subject survey discussed in Section 4.2 below provides further evidence. It was collected at the end of 2024, when COVID-19 was no longer a major public concern. In addition, it achieved a provincial distribution fairly close to that of the census, as shown in [Appendix Figure A2\(b\)](#). Given that we find robust results in the between-subject survey, the concerns of COVID-19 and regional representativeness are further reduced.

Table 3. *Heterogeneity in the Effect of the Tutoring ban on Fertility Intention.*

	Expected number of children					
	(1)	(2)	(3)	(4)	(5)	(6)
Tutoring ban	0.130*** (0.008)	0.123*** (0.009)	0.112*** (0.009)	0.110*** (0.018)	0.079*** (0.014)	0.097*** (0.007)
Ban × Female	−0.040*** (0.011)					
Ban × BA+		−0.030*** (0.011)				
Ban × Mother ≥ 40			−0.006 (0.010)			
Ban × First child is boy				−0.001 (0.022)		
Ban × Top half in education expenditure pre-policy					0.048*** (0.017)	
Ban × Top half in tutoring time pre-policy						0.027** (0.013)
Family FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	2,070	2,070	2,070	2,070	2,070	2,070

Notes: The dependent variable is the expected total number of children. Independent variables include the dummy variable for the scenario where the tutoring ban is implemented and its interaction with other group indicators. All regressions control for family fixed effects. SEs are clustered at the province level. Significance levels: *** $p < .01$, ** $p < .05$.

for mothers over age forty (column (3)).¹² It is also similar by gender of the first child, suggesting that the son preference is not a key driver here (column (4)).

Next, we explore whether families with higher pre-policy burdens of education investment and competition are more likely to increase fertility intention when the tutoring ban is implemented. Columns (5)–(6) of Table 3 show that the more money and time parents invest before the policy, the greater the increase in their fertility intention following the tutoring ban. Parents in the top half of the sample in terms of pre-policy educational expenditure or tutoring time report significantly higher fertility intentions—by 0.027–0.048 children—compared to those in the bottom half.

Our main survey focuses on parents who currently have children attending primary or middle school so that they are directly affected by the tutoring ban. However, the policy may also indirectly influence the fertility intention of married couples who do not yet have children by changing the expected educational cost for their future children. To complete this other half of the story, we further surveyed married couples currently without children.¹³ We find a similar increase of 6.5% in the expected total number of children among these couples in [Online Appendix Table A3](#), and the tutoring ban affects mainly the intensive margin of fertility intention instead of the extensive margin. This also makes our main results of parents less restrictive, because the intention to have an additional child, instead of the first child, seems to be more responsive to the tutoring ban policy.

¹² The average age of the mother in our sample is 38.8, and 94% of the mothers in the sample are under age forty-nine, the upper limit in fertility calculations (United Nations, 2024). We therefore think that most respondents in our sample are relevant in analysing the intention and potential to have an additional child, including those who are above age forty. Based on the 2015 Chinese census, which is the latest census that provides public access to micro-level data, more than 570,000 children (5.1% of total births) were born to mothers above age forty in that year.

¹³ More details on the survey sample and implementation are discussed in [Online Appendix C.2](#).

Table 4. *Effects of the Tutoring ban on Tutoring Expenditures, Time Investment, Parental Health and Perceived Educational Competition.*

Dep. var.:	Expenditure on academic tutoring (RMB/month)	Parental time on child study (hour/day)	Parent health and relationship	Perceived lower educ. competition
Dep. var. mean (no policy):	[795.3] (1)	[3.812] (2)	[0] (3)	[0] (4)
Tutoring ban	−274.551*** (55.154)	−0.690*** (0.049)	0.586*** (0.047)	0.446*** (0.042)
Family FEs & controls	Yes	Yes	Yes	Yes
N	2,028	1,936	1,968	2,070

Notes: The dependent variable is monthly spending on academic tutoring in column (1), the number of hours per day that parents spend on child study in column (2), an aggregate index (standardised) of parental health and parent-child relationship in column (3) and a dummy variable of parents perceiving other parents will reduce private tutoring post-policy in column (4). All regressions control for family fixed effects, as well as family income and the number, volatility and growth rate of local COVID-19 cases during each semester. The number of observations slightly differ across columns due to missing responses in the corresponding dependent variable. SEs are clustered at the province level. Significance level: *** $p < .01$.

3.3. *Decomposing the Change in Fertility Intention*

Couples consider multiple costs of future childrearing when making fertility decisions. These, not only include direct monetary costs such as expected education and tutoring expenditures, but also the fierce competition in child education and associated time and health costs on parents and children. To explore potential factors contributing to increased fertility intentions, we first examine whether parents' perceived education competition, as well as monetary, time and health-related costs, are affected by the tutoring ban. For perceived competition, we directly elicit parental belief that *other parents* in the same grade will reduce private tutoring after the policy. This provides useful information about perceived change in educational competition.¹⁴ For other variables, we collect detailed information in the survey on *actual* education investment behaviours for each of the two semesters before and after the implementation of the tutoring ban. We regress these measures on an indicator for the implementation of the tutoring ban, using the same family fixed-effect specification as in (1), and further control for family income and the progress of the COVID-19 pandemic in each semester to capture unobserved time differences.

Column (1) of Table 4 shows that monthly expenditure on academic tutoring significantly decreased by 275 RMB per month (or 34.5%) after the tutoring ban. Parents also spend an average 0.69 fewer hours per day supervising their children's study, as reported in column (2), representing a time saving of 18.1%.¹⁵ In addition to the decrease in monetary and time costs, we also examine whether the tutoring ban affected parental health. To measure this, we combine

¹⁴ We use this belief to measure parents' perceived level of competition, because perceived competition in our setting is typically reflected in the amount of resources (both time and money) that they think other parents and children devote to education. The more resources other parents invest, the more difficult it becomes for their own children to maintain the same academic ranking at school, which determines their admission to elite high schools and universities in China. In the between-subject survey in Section 4.2 below, we directly measure parents' perceived competition, and the correlation between these two measures is 0.671 (p -value = .000).

¹⁵ At first glance, one might expect that banning private tutoring would lead parents to take on more educational responsibilities themselves—for example, by directly tutoring their children at home—thereby increasing the time spent on supervision, which is a human capital motive. However, we observe a decline in supervision time, which can be explained by two offsetting mechanisms. First, a mechanical effect: tutoring institutions often assign homework that parents help supervise; with less tutoring, this burden naturally shrinks. Second, a competition effect: much parental effort is driven by the desire to maintain their child's relative academic ranking (Kim *et al.*, 2024). Our survey shows that, on average, parents believe that others are also reducing their educational investment, thereby lowering the effort

the survey responses on how parents perceive changes in pressure, physical health and child's closeness with both parents after the tutoring ban, rated on a ten-point scale from 'substantial deterioration' to 'no change' and then 'substantial improvement'. We then run the principal component analysis to combine these responses and use the first component as an aggregate measure in column (3) of Table 4. We find that the tutoring ban significantly increases parental health and parent-child relationships by more than half an SD.¹⁶ Lastly, in column (4), we find that nearly 45% of the parents believed that other parents will reduce private tutoring after the policy, thus suggesting that the perceived educational environment is less competitive after the tutoring ban.¹⁷

In the following, we formally quantify the relative contribution of the above potential factors to the observed higher fertility intention. To conduct the decomposition exercise, we follow the methodology proposed by Heckman *et al.* (2013) and Gelbach (2016). As a first step, we estimate a baseline regression of fertility intentions on the policy indicator while controlling for individual fixed effects:

$$Y_{is} = \tau_0 + \tau_1 Ban_s + X_{is}\tau_2 + \delta_i + \varepsilon_{is}.$$

This gives the overall effects of the tutoring ban on fertility intention $\hat{\tau}_1$. Next, we add a set of potential contributing factors M , using the values in the semester before and after the policy to match the survey scenarios without and with the tutoring ban, respectively, to form an extended specification, also with individual fixed effects:

$$Y_{is} = \alpha_0 + \alpha_1 Ban_s + \sum_j \alpha_2^j M_{is}^j + X_{is}\alpha_3 + \delta_i + \varepsilon_{is}.$$

Here M_{is}^j is the contributing factor j , specified as

$$M_{is}^j = \gamma_0^j + \gamma_1^j Ban_s + X_{is}\gamma_2^j + \delta_i + \varepsilon_{is}.$$

Gelbach (2016) showed that the total effect $\hat{\tau}_1 = \hat{\alpha}_1 + \sum_j \hat{\alpha}_2^j \hat{\gamma}_1^j$, so that the contribution of factor j is $\hat{\alpha}_2^j \hat{\gamma}_1^j$ and the remaining part is $\hat{\alpha}_1$.

We incorporate the four potential contributing factors above to decompose the observed positive effect of the tutoring ban on fertility intention:

1. monetary cost (monthly educational expenditure),
2. time cost (parental time spent on children's study),
3. health (parental health and parent-child relationships),
4. competition (whether parents perceive that other parents reduce private tutoring).

Figure 2 plots the estimated relative contributions of the four factors. Together, they can explain more than 43% of the observed increase in fertility intention. Among the explained variation,

required to maintain the same relative ranking. The observed decline suggests that the mechanical and competition effects outweigh the human capital motive.

¹⁶ The first principal component used as the dependent variable in column (3) of Table 4 is normalised to set the pre-policy value to zero as the benchmark for comparison and standardised to have an SD of one.

¹⁷ In column (4) of Table 4, the dependent variable is a dummy variable indicating whether parents perceive that other parents reduce private tutoring post-policy. It equals zero in the pre-policy period.

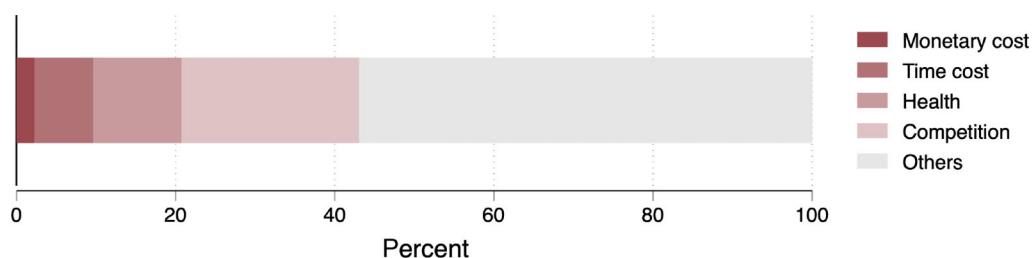


Fig. 2. *Quantifying the Contribution of Potential Mechanisms.*

Notes: The graph plots the relative contribution of potential factors to the effect of the tutoring ban on the expected total number of children, following the procedure proposed by Gelbach (2016).

perceived competition contributes the most (52%), followed by parental health (26%), while monetary and time costs account for 5% and 17%, respectively. These findings underscore the critical role of reduced competition and non-pecuniary benefits—such as improved health and parent-child relationships—relative to direct monetary or time costs in shaping fertility decisions.

4. Further Analyses

In this section, we first employ a difference-in-differences (DID) framework, leveraging city-level variation in policy intensity to strengthen the identification of the tutoring ban's effect on fertility intentions in the within-subject design. Next, we discuss a supplementary survey using a between-subject design to address potential concerns about experimenter demand effect. Lastly, we explore the short-run effect of the tutoring ban on actual fertility using data from city-level birth rates between 2016 and 2024.

4.1. Policy Intensity

In the within-subject survey, we elicit fertility intentions under the two scenarios with and without the tutoring ban. We utilise the further difference in the policy intensity of the tutoring ban across cities to construct a continuous DID model. Below we present the model specification and discuss the policy intensity measure in detail. The DID regression is specified as

$$Y_{ics} = \beta_0 + \beta_1 Ban_s + \beta_2 Ban_s \times Intensity_c + \delta_i + \varepsilon_{ics},$$

where Y_{ics} is the fertility intention of family i in city c under scenario s ; Ban_s is a binary indicator for implementing the tutoring ban in scenario s , and it is further interacted with $Intensity_c$, which is the intensity of the policy shock in city c ; δ_i is the family fixed effect and ε_{ics} is the residual term.

The city-level policy intensity, $Intensity_c$, is constructed by a presumably exogenous measure: the density of tutoring institutions (number per capita) in each city, derived from administrative firm registration records five years prior to the policy. The key intuition is that, when the tutoring ban was abruptly implemented nationwide, cities with a higher pre-policy density of tutoring institutions would experience a more substantial disruption, making the policy impact more 'intense'. This approach aligns with a well-established literature that leverages differential

Table 5. *Fertility Intention Change by Intensity of the Tutoring Ban.*

	Academic tutoring time (hour/week)	Expected number of children
Dep. var.:	[2.232]	[1.610]
Dep. var. mean (no policy):	(1)	(2)
Tutoring ban	−0.446*** (0.084)	0.083*** (0.010)
Tutoring ban × Intensity	−0.104* (0.055)	0.015** (0.005)
Family FEs	Yes	Yes
COVID-19 controls	Yes	Yes
N	2,012	2,012

Notes: ‘Intensity’ is measured by the number of tutoring institutions per capita in a city in 2016 (five years prior to the tutoring ban) and is standardised to have zero mean and unit variance. SEs are clustered at the province level. Significance levels: *** $p < .01$, ** $p < .05$, * $p < .1$.

exposure to a common shock using the shift-share design (Autor *et al.*, 2013; Borusyak *et al.*, 2025).¹⁸

To facilitate the interpretation of regression coefficients, we standardise the density of tutoring institutions to have a mean of zero and a variance of one. To validate this intensity measure, we first confirm a significantly larger drop in academic tutoring time following the tutoring ban in cities with a higher pre-policy density of tutoring institutions, as shown in column (1) of Table 5. Weekly tutoring time significantly decreased by 0.446 hours after the policy in cities with average density (the coefficient of ‘*Tutoring ban*’, where ‘*Intensity*’ is at mean zero), and with one-SD higher policy intensity, tutoring time decreased by an additional 0.104 hours. Huang *et al.* (2025) also found a larger decrease in job postings and a higher number of firm exits in the tutoring industry after the ban in cities with higher pre-policy density using the same measure. These findings suggest that cities with a higher density of tutoring institutions prior to the policy indeed experienced a larger shock to tutoring activities.

We present the change in fertility intention by the intensity of the policy shock in column (2) of Table 5. Our estimates show that in cities with an average density, the implementation of the tutoring ban significantly increases the fertility intention by 0.083 children (the coefficient of ‘*Tutoring ban*’, where ‘*Intensity*’ is at mean zero). For cities with one-SD higher policy intensity, fertility intention significantly increases by an additional 0.015 children. These consistent results provide robust support for the finding that the implementation of the tutoring ban indeed improves parents’ fertility intentions.¹⁹

4.2. *Alternative Survey Design*

The within-subject design used in our main survey has the advantage of effectively controlling for unobserved heterogeneity through individual fixed effects, but may be biased due to the

¹⁸ Given the plausibly exogenous intensity of the policy exposure, our continuous DID analysis is less prone to the selection bias, as discussed by Callaway *et al.* (2024). Nonetheless, we acknowledge that ‘strong parallel trends’ (Callaway *et al.*, 2024) are assumed to be satisfied.

¹⁹ We further look at the companion sample of couples without children in Online Appendix Table A4. Compared to the parent sample, the tutoring ban has a smaller positive effect in the average city; however, for each SD increase in policy strength, fertility intention increases more substantially—by 0.035 children. This could be because couples without children do not pay the same attention to education policies as parents, so their fertility intention is affected only when the policy intensity is high enough.

Table 6. *Between-Subject Survey.*

Dep. var.:	Expected number of children	Perceived educ competition (0–10)
Dep. var. mean:	[1.580] (1)	[8.263] (2)
Tutoring ban	0.127*** (0.037)	–3.660*** (0.215)
Family controls	Yes	Yes
Province FEs	Yes	Yes
N	1,336	1,336

Notes: Family controls include the mother’s age, log of family income, respondent’s gender, parents’ years of education, Hukou status, marital status, and the first child’s age and gender. SEs clustered at the provincial level. Significance level: *** $p < .01$.

experimenter demand effect (Fuster and Zafar, 2023). We addressed this concern by running another round of the parent survey at the end of 2024 using the between-subject design, where the experimenter demand effect should in general be weaker. Each respondent was randomly presented with one scenario only, either with or without the tutoring ban.²⁰ We used the survey panel managed by Credamo, a leading professional research and survey company in China, and distributed the survey to its subject pool.²¹ We obtained 1,348 responses from parents with at least one child currently enrolled in primary or middle school. These respondents were spread across 216 cities in thirty provinces, and their regional distribution generally aligns with the actual population. [Online Appendix C.1](#) provides more details on the survey design, sample characteristics and balance check of randomisation.

We compare the main effect of the tutoring ban on fertility intentions in column (1) of Table 6 between two survey designs. We find that the expected number of children significantly increases by 0.127 (or 8%) if the tutoring ban is implemented using the between-subject survey, a magnitude closely aligned with the 0.109 (7%) increase observed in the original within-subject survey, reinforcing the robustness of our findings across different survey designs and distribution methods. The fact that we find similar policy impacts on intended fertility in both the within- and between-subject designs also suggests that our original results are unlikely to be driven by experimenter demand effects.²²

The Credamo panel survey also helps address concerns about potential self-selection bias in the within-subject survey on the national education platform. Participants were unaware the

²⁰ In order to keep comparability with the within-subject survey, we maintain a similar structure in the between-subject survey, in which we first introduce the policy, and then elicit fertility intention and educational investment behaviours under the randomly presented policy scenario. See the complete questionnaire in [Online Appendix D.2](#).

²¹ We first return to the nationwide education platform for our between-subject survey. Unfortunately, the user base of the national educational cloud platform (where we collected the initial survey data) had significantly declined, as many local educational authorities discontinued their subscriptions after the COVID-19 pandemic ended and schools resumed in-person teaching. As a result, we received only 274 valid responses over a two-week distribution period at the end of 2024, covering fourteen cities across nine provinces (compared to 136 cities in twenty-eight of the thirty-one Chinese provinces in our initial survey). Nevertheless, we find consistent and similar-sized results using this smaller sample.

²² Admittedly, if participants were highly sophisticated and inferred that differential responses were expected across the policy and no-policy scenarios even in a between-subject setting, such effects could theoretically bias the results. However, we think that the experimental demand effect is unlikely to be a major factor here for the following two reasons. First, if the experimental demand effect were important, we would expect the estimated effect to be larger in the within-subject design than in the between-subject design. In contrast, we find that the estimated effects are very similar across the two designs, with the between-subject estimate even slightly larger. Second, our difference-in-differences analysis on actual birth rates in Section 4.3 below shows that regions with higher policy intensity experienced significantly greater increases in actual birth rates relative to regions with average intensity. The correspondence between stated intentions and actual behaviour further suggests that our survey findings are not merely an artefact of experimental demand.

survey focused on the tutoring ban or fertility intentions when they joined, as they saw only a generic title ('Survey Questionnaire') and were offered a fixed payment of 20 CNY (≈ 3 USD). Questions about the tutoring ban and fertility appeared midway through the survey, after screening and demographic questions, and participants who exited early received no compensation. This design ensures that selection into the survey is unlikely to correlate with the tutoring ban or child education. The robustness of our findings in this between-subject survey suggests that selection bias related to parents' policy sensitivity in the main survey is unlikely to drive our results.

In addition, because the between-subject survey was conducted at the end of 2024 when COVID-19 was no longer a public agenda of the society, this additional result will not be affected by the pandemic. In [Online Appendix Table A5](#), we consistently find that the elicited total educational expenditure, tutoring expenditure and time are significantly lower under the scenario of the tutoring ban, which further supports the conclusion that the reduction in educational investment after the tutoring ban was not likely driven by COVID-19.

Lastly, in the supplementary between-subject survey, we directly measure parental beliefs about the competitiveness of the educational environment under the tutoring ban. This approach provides a potentially more direct measure of parents' perceived competitiveness compared to the within-subject survey, which relied on beliefs about whether other parents would reduce private tutoring. Parents rated this on a ten-point scale, ranging from 'no competition' to 'extremely competitive'. As shown in column (2) of [Table 6](#), the average perceived competition level is high at 8.263 in the absence of the policy. However, under the tutoring ban, it drops significantly by 3.660 points—a 44% reduction—bringing the average perception below the midpoint of 5. This provides more direct evidence that parents perceive the educational environment as significantly less competitive following the tutoring ban, further supporting the conclusion from the decomposition analysis that the reduction in perceived education competition contributes significantly to the increase in fertility intentions.²³

4.3. Actual Birth Rates

While our study focuses on fertility intentions, we have the opportunity to explore the policy impact on actual fertility after three years of implementation. We collect city-level birth rate data both before and after the tutoring ban and conduct the DID analysis using the same policy intensity measure as in the analysis of fertility intentions in [Section 4.1](#).

Annual data on fertility at the city level are not systematically available in China. For the period prior to the tutoring ban in 2021, we rely on the 2020 Chinese census. The 2020 census provides the full population count in each city for age 0 (born in 2020), for the 1–4 age bin and five-year bins for older age groups. We use the group of age 0 to directly get the number of births in 2020, and use the 1–4 age bin to calculate the average number of births per year between 2016 and 2019. For the period after the policy in 2021, we could not find fertility data at the city level for the years during the COVID-19 pandemic. But, since 2023, Chinese city governments started to systematically publish the birth statistic in their annual Statistical Communiqué. We were able to find these data for 319 out of the 337 Chinese cities for the year 2023, and for 204 cities for

²³ It is not feasible to perform the same decomposition exercise in the between-subject design because such a decomposition suffers from endogeneity issues due to the inability to control for individual fixed effects. In this design, unobserved characteristics may correlate with both the fertility intention and the contributing factor, even if the policy scenario is randomly assigned. In contrast, the within-subject design mitigates this problem by directly controlling for individual fixed effects.

Table 7. *Tutoring Ban and Actual Birth Rates.*

Dep. var.: Dep. var. mean (pre-policy):	Birth rate (‰) [10.208]	
	(1)	(2)
Tutoring ban × Intensity	0.261*** (0.078)	0.270*** (0.078)
City FEs	Yes	Yes
Year FEs	Yes	Yes
City-year controls	Yes	Yes
Sample	Full sample	Exclude 2024
N	1,012	841

Notes: Data from the China Census (2020) and city-level Statistical Communiqué (2023; 2024). The dependent variable is the number of births per 1,000 population in each city, and is available for the 2016–19 average, 2020, 2023 and 2024. ‘Intensity’ is measured by the number of tutoring institutions per capita in a city in 2016 (five years prior to the tutoring ban) and is standardised to have zero mean and unit variance. City-year control variables (City Statistical Yearbook, 2020) include the lagged GDP per capita, housing price, per-student public educational expenditure and the number of COVID-19 cases. SEs clustered at the city level. Significance level: *** $p < .01$.

the year 2024.²⁴ Using these data, we calculate the city-level annual birth rate (number of births per 1,000 population) for the 2016–19 average (treated as a single year as the 2020 census only provides the total number for ages 1–4), 2020, 2023 and 2024.²⁵

Using these data, we conduct the same DID analysis as for the intention data to examine city-level trends in birth rates before and after the policy, comparing cities with varying levels of policy intensity. Policy intensity is measured by the per capita number of tutoring institutions in 2016, based on administrative firm registration records from five years before the policy. This comparison helps shed light on the potential effects of the policy on actual fertility behaviour. The analysis excludes fertility data from years prior to 2016 to avoid concerns about reverse causality. The DID specification is

$$Y_{ct} = \beta_0 + \beta_1 Ban_t \times Intensity_c + \beta_2 X_{ct} + \delta_c + \gamma_t + \varepsilon_{ct},$$

where Y_{ct} is the birth rate in city c in year t , Ban_t is the policy indicator that equals 1 for t after 2021 and $Intensity_c$ is the standardised policy intensity in city c (with a mean of zero and a variance of one across cities). In addition, X_{ct} is a set of city-level time-varying control variables, including the lagged GDP per capita, housing price, per-student public educational expenditure, and the number of COVID-19 cases. Here δ_c and γ_t are city and year fixed effects, and ε_{ct} is the error term. SEs are clustered at the city level.

The results are reported in Table 7. The average birth rate was 10.208‰ before the policy (during 2016–20) and decreased to 6.504‰ in 2023. However, under this overall trend, we find that the tutoring ban has slowed the fertility decline after its implementation in 2021. In column (1) that uses the full sample, when the policy intensity is one SD above the mean, the city’s birth rate significantly increases by 0.261 (in ‰) relative to the city with the average intensity (when ‘Intensity’ is at mean zero) after the tutoring ban was implemented. This estimated marginal

²⁴ The City Statistical Communiqué for the statistics of 2024 was gradually released during 2025. As of June 2025, 204 cities had published their Communiqué.

²⁵ Online Appendix Table A6 reports the summary statistics. It worth noting that the TFR, which is more closely linked to the fertility intention (expected total number of children) that we estimate, would be difficult to calculate without the micro-level data on women’s ages and birth records in each city, which are not publicly available for the 2020 census or more recent years.

effect of the one-SD higher policy shock translates to a 7.0% slowdown of the birth rate decline between the 2016–20 average and 2023. In column (2), we further exclude the year 2024, which has data on only two-thirds of cities, and get a very similar estimate. We further explore the dynamic effect and do not find a clear pre-trend in [Online Appendix Figure A3](#). This finding of the policy impact on actual birth rates is broadly consistent with the higher fertility intentions that we elicited from the survey data.²⁶

5. Concluding Remarks

In this paper, we investigate whether alleviating competition-driven tutoring can increase fertility intentions. Our study exploits China's private tutoring ban implemented in 2021 as a natural experiment. For the main analysis, we conducted a nationwide within-subject survey of primary and middle school parents and elicited subjective beliefs of having additional children. Using both the individual fixed-effect model and the difference-in-differences framework, we find a significant increase in expected total fertility of 7% (from 1.61 to 1.73 children per family) when the tutoring ban is implemented. This result is robust in a complementary between-subject survey, which alleviates concerns about the experimenter demand effect and also shows an 8% increase in the expected total fertility. Further analysis reveals that perceived lower educational competition among parents contributes the most to this increase in fertility intentions, followed by substantially improved parental health and parent-child relationships, and then alleviated time and financial burdens. Lastly, although the overall fertility is declining rapidly in China, we find that the actual birth rates are declining significantly less in regions more exposed to the policy three years after the implementation relative to those less exposed, consistent with the elicited fertility intentions.

Whether reduced competition and lower household investment in education decrease the child generation's human capital is worth further research. The total loss of human capital is potentially limited if part of the learning through private tutoring reflects testing skills instead of productive human capital. Child quality also extends beyond cognitive skills to include important aspects such as physical and mental health, which are substantially deteriorated under intense competition in education, as shown by various studies. In our survey, we found that sleep time increased significantly by 0.59 hours per day after the tutoring ban, and that the parent-child relationship improved by more than half an SD. If these health-related burdens can be alleviated, overall child quality may improve. Future work on these aspects could provide a more complete evaluation of the quantity-quality trade-off. In addition, we acknowledge that the private tutoring ban in China may generate other impacts, such as shifting capital and employment away from the tutoring industry, and an overall cost-benefit evaluation of this policy is left for future research.

In general, the private tutoring ban in China offers a promising example of how reducing the pressure of educational competition can positively influence fertility intentions and actual childbirth choices. As countries grapple with declining fertility rates, addressing the role of edu-

²⁶ We acknowledge that these two sets of results are not fully quantitatively comparable due to different fertility calculations. The actual fertility is measured by the annual birth rate of a cross-sectional population, while the fertility intention is the expected total fertility over the life cycle of one cohort, whose actual data at the city-by-year level are not available as of now. We leave the quantitative evaluation of how much of the increased intention translates to actual fertility for future research.

cational competition and its broader societal impacts could be important in developing effective and sustainable pro-natal policies.

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